Logistic Distribution

1 Definition

A continuous random variable X is said to have a Logistic distribution if its CDF is given by,

$$F_X(x) = \frac{e^{\frac{x-\alpha}{\beta}}}{1+e^{\frac{x-\alpha}{\beta}}} = \frac{1}{1+e^{-\left(\frac{x-\alpha}{\beta}\right)}},$$

where $\alpha \in \mathcal{R}, \beta > 0$.

$$f_X(x) = \frac{\frac{1}{\beta}e^{-\left(\frac{x-\alpha}{\beta}\right)}}{\left\{1+e^{-\left(\frac{x-\alpha}{\beta}\right)}\right\}^2} = \frac{\frac{1}{\beta}e^{\frac{x-\alpha}{\beta}}}{\left(1+e^{\frac{x-\alpha}{\beta}}\right)^2}, x \in \mathbb{R}$$

Note that, $f_X(\alpha - h) = f_X(\alpha + h)$

2 Moments

Mean = Median = Mode = α. The distribution is symmetric about α.

3 Standard Logistic Distribution

If $X \sim \text{Logistic}(\alpha, \beta)$, $Y = \dfrac{X-\alpha}{\beta} \sim \text{Logistic}(0,1)$.

$F_Y(y) = P[Y \leqslant y] = P[X < \alpha + \beta y] = F_X(\alpha + \beta y) = \dfrac{e^y}{1+e^y}$ or $\dfrac{1}{1+e^{-y}}$

Clearly, $Y \sim$ Logistic $(0,1)$.

Remark:

If $X \sim$ Logistic (α, β),

$$F_X(x) = \frac{e^{(x-\alpha)/\beta}}{1+e^{(x-\alpha)/\beta}} \quad x \in \mathbb{R}.$$

$$\Rightarrow \quad 1 - F_X(x) = \frac{1}{1+e^{\frac{x-\alpha}{\beta}}}$$

$$\Rightarrow \quad \frac{F_X(x)}{1-F_X(x)} = e^{\frac{x-\alpha}{\beta}} \Rightarrow \ln\left(\frac{F_X(x)}{1-f_X(x)}\right) = \frac{x}{\beta} - \frac{\alpha}{\beta}. \text{ which is linear in } x.$$

The log-odds or logit is linear. This fact is used for logistic regression for Binary Distribution.

4 Problem

The unit interval $(0, 1)$ is divided into two sub-intervals by picking up a point at Random, from inside the interval, denoting Y and Z the length of longer and the shorter sub-interval. Show that $E\left(\frac{Y}{Z}\right)$ does not exist.

Let X be the point selected randomly from the interval, $X \sim U(0, 1)$ and

$$Y = \max(X, 1 - X); Z = \min(X, 1 - X)$$

$$\begin{aligned} E\left(\frac{Y}{Z}\right) &= \int_0^1 \frac{\max(x, 1-x)}{\min(x, 1-x)} f_x(x) dx \\ &= \int_0^1 \frac{\max(x, 1-x)}{\min(x, 1-x)} dx \\ &= 2\int_0^{1/2} \frac{1-x}{x} dx \\ &= 2\left\{\lim_{a\to 0+} \int_a^{1/2} \frac{dx}{x} - \int_0^{1/2} dx\right\} \\ &= 2\left\{\lim_{a\to 0+} (\ln(1/2) - \ln a) - 1/2\right\} \\ &\to +\infty \end{aligned}$$

So, $E\left(\frac{Y}{Z}\right)$ does not exist.